

# Day 06: Transforming SLR models

Name: \_\_\_\_\_

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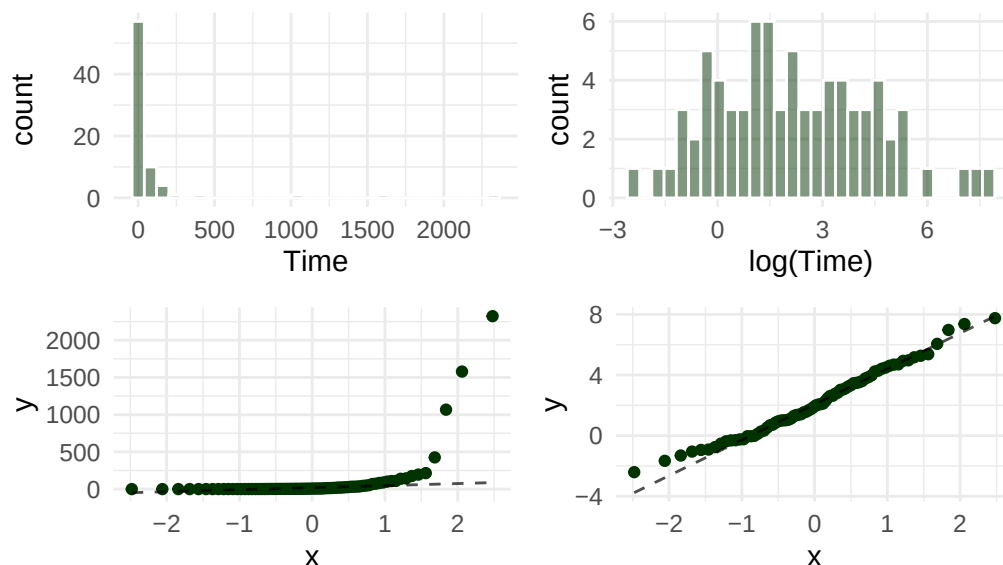
Last time -- and today

We checked: are any conditions for SLR *violated*? Is there a strong predictive relationship between X and Y? **Today:** how can we fix the worst violation?

If the linearity condition is not met, we get systematically biased predictions, might draw false conclusions about the existence or strength of a relationship, and all inference about  $\beta_0, \beta_1, \hat{\mu}(Y | X)$ ,  $\text{Pred}(Y | X)$  becomes meaningless.

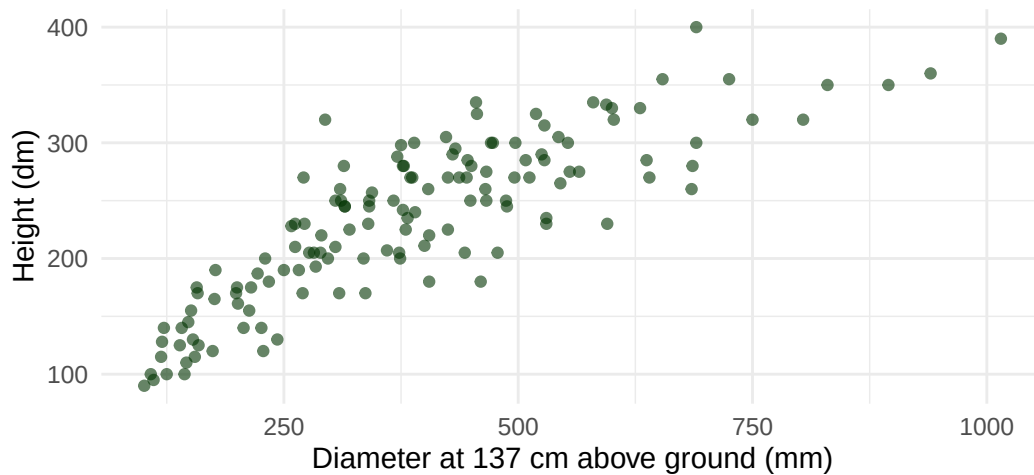
Easing skew

If a set of data values is skewed to the right, taking the (natural) log of each value *can* result in a data set that is roughly symmetric and often roughly normal.

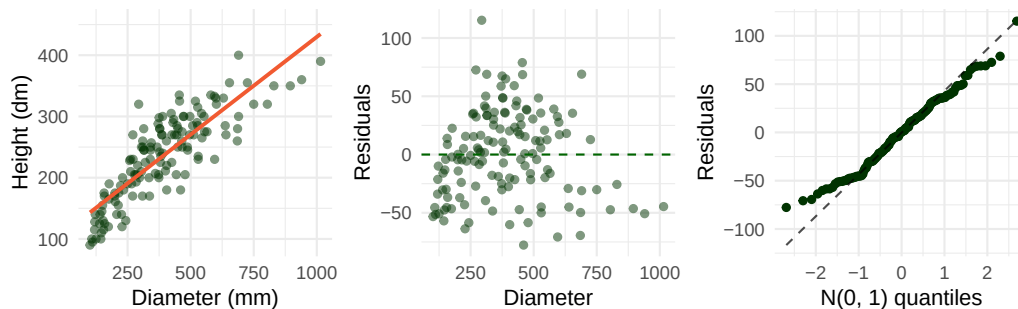


Data for today: red cedar

How are tree height and tree diameter related for the western red cedar?



Are the conditions violated?

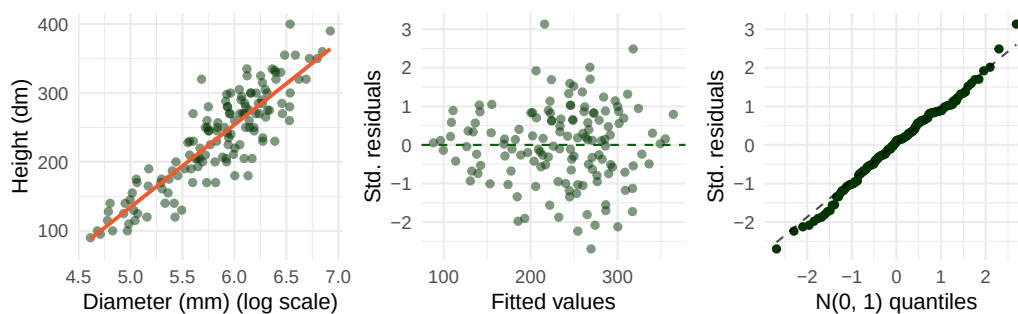


Adding `gf_smooth()` to the residual plot can help reveal curvature – but trust your eyes first, and don't over-interpret small samples.

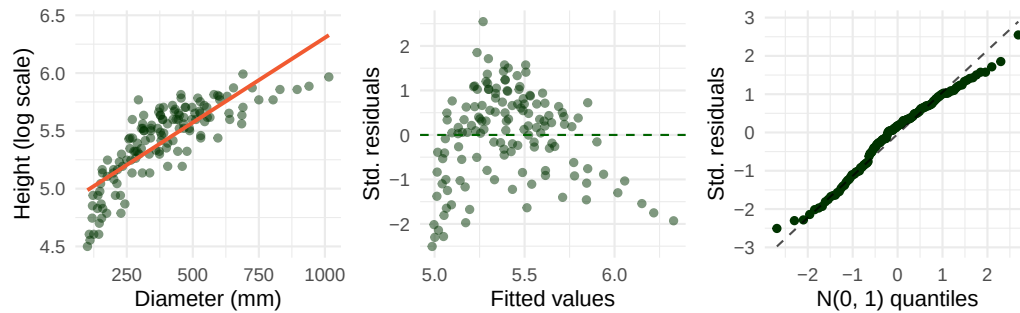
Which condition(s) look violated here: (a) linearity (b) independent errors (c) normal errors (d) equal variance (e) outliers (f) none?

**Answer:** \_\_\_\_\_

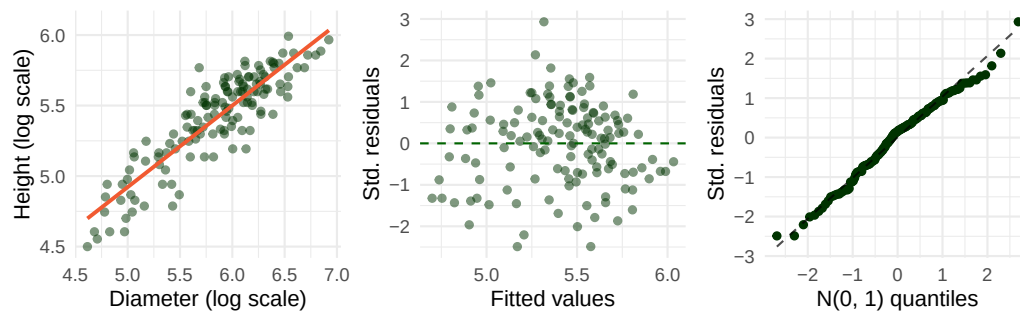
Does transforming X help?



Does transforming Y help?



Transforming both X and Y?



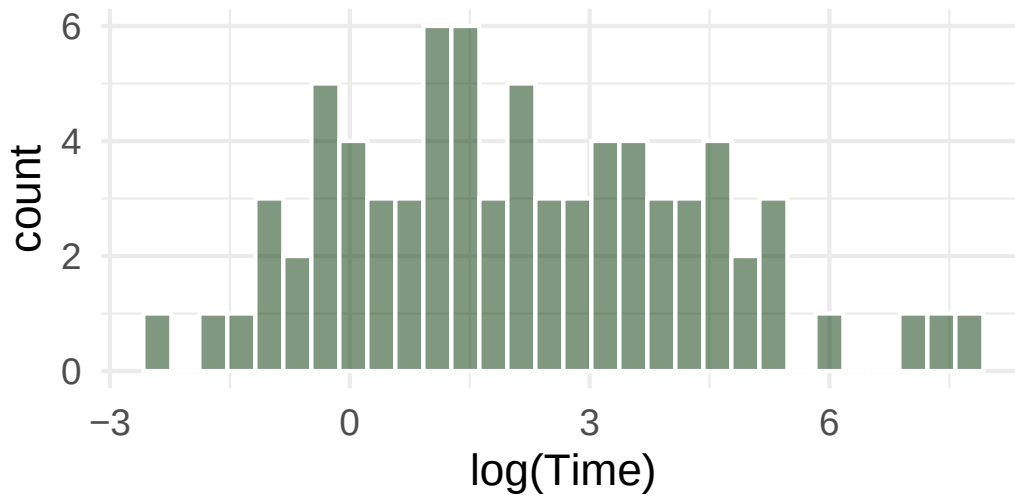
#### 💡 Group task

Work through the first example on the handout in your groups (online version with R chunks on Moodle). Compare the three transformation options above – which one best resolves the condition violation(s) you flagged earlier?

## Back-transforming

Converting a transformed variable back to its original scale.

Example: back-transforming a mean



Mean of  $\log(\text{Time}) = 2.146$ .

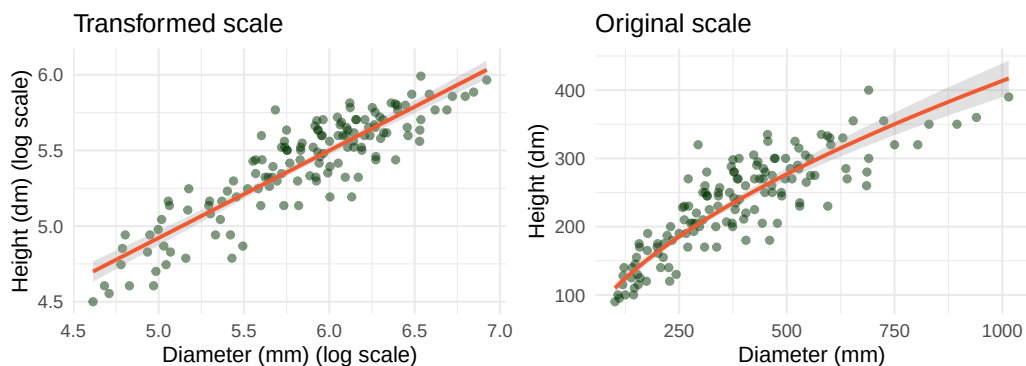
Back-transformed mean:  $e^{2.146} \approx 8.55$ .

#### R note

$\log()$  is the *natural* log;  $\exp(x)$  calculates  $e^x$ . **Note:** back-transforming the mean of the logs does **not** generally equal the mean on the original scale (98.56) – it estimates the geometric mean/median instead.

Displaying a transformed model

`gf_lm(formula = log(y) ~ log(x), backtrans = exp, interval = "confidence")` plots the fitted line and CI back-transformed onto the *original* scale, so you can show results in units your audience understands.

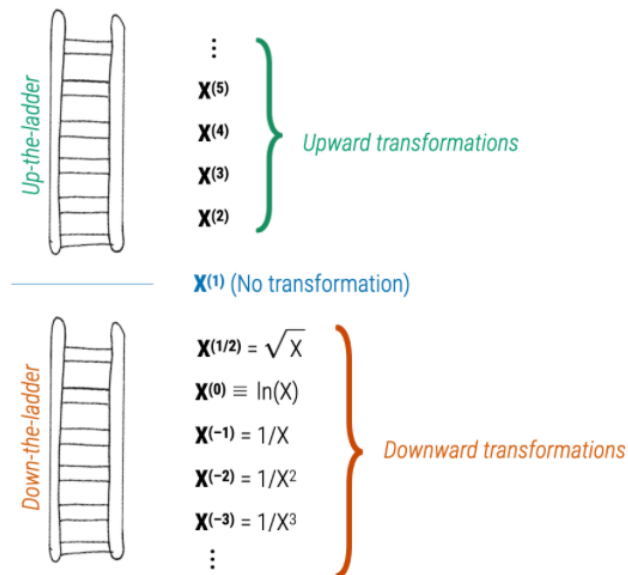


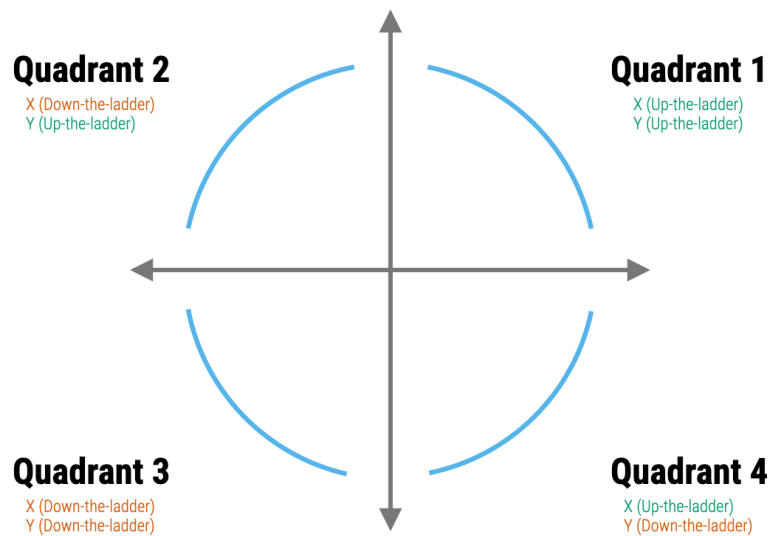
## Rules of thumb

- **Transform X:** if the mean function is nonlinear, but monotonic, and the residual variance is constant.
- **Transform Y:** if the mean function is nonlinear *and* residual variance increases as the mean increases (log, reciprocal, or square root often work).
- **Log rule:** if values range over more than 1 order of magnitude and are strictly positive, the natural log is likely helpful.
- **Range rule:** if the range is considerably less than 1 order of magnitude, transformations are unlikely to help.
- **Square roots** are useful for count data.

## Ladder of transformations & Rule of the Bulge

Tukey and Mosteller's "Rule of the Bulge" gives a visual guide: match the direction your scatterplot bulges to the ladder position, and move X and/or Y toward that end to straighten the relationship.

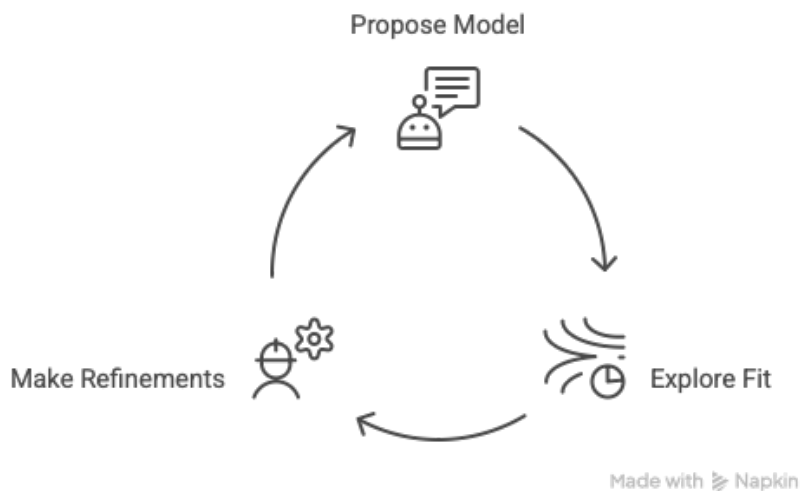




#### Note

Image credit: Andrew Zieffler

Modeling is an iterative process



Issues with transformations

- You're often guessing – statistics is an art *and* a science.
- Transformations change the \_\_\_\_\_ of the parameters – you need to back-transform to give interpretable results.

- Transformations change the SEs of the parameters.
- Not always easy to keep track of all your assumptions once you've transformed.